

Robust Image-Based Visual Servoing Formation Control for Quadrotors Without Communication via Reinforcement Learning

Xinning Yi[✉], Hao Liu[✉], *Senior Member, IEEE*, Haibin Duan[✉], *Senior Member, IEEE*,
Biao Luo[✉], *Senior Member, IEEE*, and Shan Zuo[✉], *Member, IEEE*

Abstract—The image-based visual servoing (IBVS) formation control problem for quadrotors in global navigation satellite system (GNSS)-denied and communication-degraded environments is addressed via reinforcement learning (RL) algorithms. A robust leader–follower formation control method is proposed to achieve formation control, and the follower quadrotors track the leader feature using the virtual camera technique and sphere-based image moments. In order to overcome the absence of intervehicle communication, a leader observer is developed to estimate the state of the leader quadrotor. For the leader–follower IBVS dynamics involving unknown uncertainties and external disturbances, robust visual servoing and attitude control laws are proposed using off-policy RL algorithms without relying on precise dynamical models. The effectiveness of the proposed control scheme is validated through both theoretical analysis and case study results.

Index Terms—Formation control, quadrotors, reinforcement learning (RL), robust control, visual servoing.

I. INTRODUCTION

THE coordination of unmanned aerial vehicles (UAVs) has become a prominent research topic in recent years, facilitating efficient and precise multiagent operations in complex and dynamic environments [1], [2], [3]. Formation control of quadrotors constitutes a fundamental component of UAV coordination, ensuring synchronized and effective multiagent operations in challenging scenarios such as search and rescue, inspection and maintenance, and environmental monitoring [4], [5]. The leader–follower approach is widely used in quadrotor formation control, where the leader follows a desired trajectory and the followers maintain a prescribed offset to

track the leader’s position [6], [7]. In common implementations of quadrotor formation control using the leader–follower structure, position information and intervehicle communication are required to maintain relative positions and achieve information exchange within the formation [8]. However, satellite signals and communication networks are vulnerable to interference or blockage in environments with electromagnetic disturbances or physical obstructions, making the global navigation satellite system (GNSS) and communication easily unavailable [9], [10], [11]. Therefore, achieving formation control becomes significantly challenging in GNSS-denied environments without communication [12], [13].

As a low-power and cost-effective sensor with rich informational capacity, camera provides an effective approach for the quadrotor control in GNSS-denied environments, known as visual servoing control [14]. Image-based visual servoing (IBVS) utilizes image feature information obtained from onboard cameras to achieve robotic motion control without relying on position information [15]. In [16], a visual observer was designed to estimate relative formation tracking errors, enabling leader–follower formation tracking of mobile robots. In [17], an optimal visual servoing-based formation control scheme was presented for multiple wheeled mobile robots. In [18], a visual servoing swarming mechanism was developed for surface vehicles. However, these studies primarily focus on ground or surface platforms. To achieve IBVS quadrotor formation control using the leader–follower structure, it is essential to track aerial moving targets. In current works related to IBVS tracking control for quadrotors, the focus has mainly been on 2-D planar target tracking, primarily addressing tasks such as hovering and ground target tracking using downward-facing cameras [19], [20], [21]. The change of the tracked target to an aerial moving target makes the selection and extraction of 3-D dynamic features a key challenge. Several works were presented to explore the quadrotor interception control methods for tracking and intercepting aerial moving targets [22], [23], [24]. In [22] and [23], image point features were used to represent the UAV target, but the projection mapping caused the loss of depth information, resulting in insufficient representation of the target. In contrast, image moment features provide a more comprehensive description of the target, offering increased robustness and aiding in depth estimation, thereby enhancing the advantages in aerial target tracking tasks [24].

Received 19 January 2025; revised 29 August 2025 and 5 January 2026; accepted 13 February 2026. Date of publication 26 February 2026; date of current version 2 September 2026. This work was supported in part by the National Natural Science Foundation of China under Grant 62273015, Grant U23B2032, Grant 62350048, Grant 62373375, and Grant U24B20156; and in part by the Science and Technology Innovation Program of Hunan Province under Grant 2024RC1011. (Corresponding author: Hao Liu.)

Xinning Yi and Hao Liu are with the Institute of Artificial Intelligence, Beihang University, Beijing 100191, China (e-mail: yixinning@buaa.edu.cn; liuhao13@buaa.edu.cn).

Haibin Duan is with the School of Automation Science and Electrical Engineering, Beihang University, Beijing 100191, China (e-mail: hbd-uan@buaa.edu.cn).

Biao Luo is with the School of Automation, Central South University, Changsha 410083, China (e-mail: biao.luo@csu.edu.cn).

Shan Zuo is with the Department of Electrical and Computer Engineering, University of Connecticut, Storrs, CT 06269 USA (e-mail: shan.zuo@uconn.edu).

Digital Object Identifier 10.1109/TNNLS.2026.3665724

Degraded communication prevents the followers directly obtaining information from the leader, causing significant difficulties in maintaining the formation structure [25], [26], [27]. For the unknown leader state information in the formation control, developing adaptive controllers and observer estimation schemes can be used to address the issue [28]. In [29], the leader was considered to have unknown but bounded control inputs, and an adaptive dynamic event-triggered protocol was designed to ensure output consensus among agents. In [30], an adaptive control strategy was presented to achieve the formation control without the feature depth and velocity information of leaders. In [31], a real-time observer was developed to estimate the unknown intrinsic and extrinsic camera parameters and unknown coefficients of the feature point. Besides, current works on IBVS formation control for quadrotors, such as [25] and [32], often rely on simplified models or assumptions of constant altitude and speed to maintain the formation structure. Therefore, developing IBVS formation control methods for six-degree-of-freedom quadrotor models in communication-degraded environments provides greater generalization capabilities and a broader range of application scenarios.

External disturbances and model uncertainties in quadrotor formation control systems can hardly be neglected in complex denied environments, making the design of robust control laws without precise model essentials. Reinforcement learning (RL) provides an effective control approach, enabling optimal control in complex and uncertain dynamic systems [33]. For the nonlinear systems involving unknown uncertainties, system identification and off-policy RL are commonly used to derive the robust model-free control. In [34] and [35], identifier-critic network architectures were presented to relax system dynamics constraints and enable decentralized control. In [36] and [37], H_∞ control schemes via the off-policy RL algorithm were presented for completely unknown continuous-time systems. In recent years, RL-based control methods have gained increasing popularity in quadrotor applications, providing promising solutions to complex and high-dimensional control tasks. In [38], an optimal control policy was presented for underactuated vertical take-off and landing aerial vehicles subject to the unknown mass and inertia matrix. In [39], a robust optimal 3-D cooperative guidance law was developed for intelligent aerial vehicles to intercept unknown maneuvering targets in the presence of external disturbances. In [40] and [41], model-free robust formation control methods were developed for cooperative underactuated quadrotors with unknown dynamics and disturbances. In [6], an RL-based fault-tolerant formation control scheme was presented for networked fixed-wing UAVs to handle actuator faults and external disturbances. However, the above studies rely on direct state measurements or inter-vehicle communications and do not consider vision-based feedback mechanisms. Consequently, achieving robust model-free control for quadrotor visual servoing formation systems under unknown uncertainties remains a crucial and challenging problem.

In this article, the robust IBVS formation control problem for quadrotors in GNSS-denied and communication-degraded environments is addressed. The leader-follower image error

dynamics are formulated to model the relative motion between the leader and follower quadrotors using image features. A leader observer is introduced to estimate the state of the leader quadrotor. Robust visual servoing and attitude control laws are then proposed using off-policy RL algorithms to address uncertainties in the quadrotor formation control system. The main contributions are shown in the following aspects.

First, the problem of feature extraction and visual servoing tracking of aerial moving targets in GNSS-denied environments is addressed. Different from [38], [39], [40], and [41], the proposed leader-follower model for quadrotors based on visual feedback in the current paper enables aerial moving target tracking without relying on direct position information measurement.

Second, the IBVS formation control problem for quadrotors without relying on the intervehicle communication or strict scenario assumptions is addressed by estimating the unknown leader state and incorporating the full quadrotor dynamics. However, the leader states were considered to be known in [23], and the altitudes and speeds of UAVs were assumed to be constants in [32].

Third, a robust visual servoing control law is proposed for the quadrotor formation control system involving unknown uncertainties. Compared to [30] and [42], the proposed off-policy RL-based control method demonstrates robustness against external disturbances and reduces sensitivity to model inaccuracies in the visual servoing formation control system for quadrotors.

The remaining parts of this article are organized as follows. In Section II, leader-follower image error dynamics of quadrotors are derived, and the problem formulation is introduced. The leader observer, visual servoing control law, and attitude control law for the IBVS formation control system of quadrotors are designed in Section III. The case study results are shown in Section IV, and conclusions are given in Section V.

Notations: Denote $sk(\cdot)$ as the skew-symmetric matrix, $\times$ as the cross product, $I_n \in \mathbb{R}^{n \times n}$ as an identity matrix, and $0_{m \times n} \in \mathbb{R}^{m \times n}$ as a zeros matrix. Let ∇ denote the gradient operator, and one has that $\nabla f(x) = \partial f / \partial x$.

II. PROBLEM FORMULATION

A. Leader-Follower Dynamics

Let ${}^e\mathcal{F} = \{{}^eX, {}^eY, {}^eZ\}$ be the inertial frame and ${}^b\mathcal{F}_j = \{{}^bX_j, {}^bY_j, {}^bZ_j\}$ the body frame of the quadrotor j . The relationship between the two frames is described by the rotation matrix R_j . The position and linear velocity of the quadrotor j are denoted as $p_j = [p_{xj}, p_{yj}, p_{zj}]^T \in \mathbb{R}^3$ and $v_j = [v_{xj}, v_{yj}, v_{zj}]^T \in \mathbb{R}^3$ in ${}^e\mathcal{F}$, respectively. The Euler angles and angular velocity of the quadrotor j are represented by $\Theta_j = [\phi_j, \theta_j, \psi_j]^T \in \mathbb{R}^3$ and $\omega_j = [\omega_{xj}, \omega_{yj}, \omega_{zj}]^T \in \mathbb{R}^3$, denoting the roll, pitch, and yaw motion, respectively. The dynamics of the quadrotor j can be given by

$$\begin{aligned} \dot{p}_j &= v_j \\ m_j \dot{v}_j &= -f_{Tj} R_j e_3 + m_j g e_3 + d_{fj} \\ \dot{R}_j &= R_j sk(\omega_j) \\ J_j \dot{\omega}_j &= -\omega_j \times J_j \omega_j + f_{\tau j} + d_{\tau j} \end{aligned} \quad (1)$$

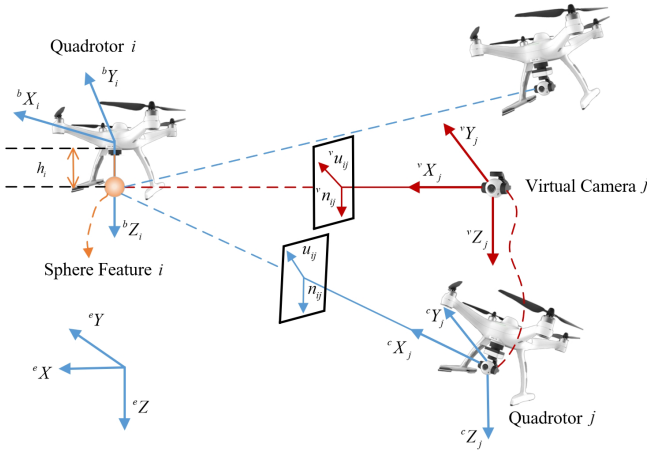


Fig. 1. Leader-follower model based on visual feedback.

where m_j and J_j are the total mass and symmetric inertia matrix. The thrust force and torque inputs of the quadrotor j are denoted as f_{Tj} and τ_j , and the uncertainties related to model uncertainties and external disturbances are represented by d_{fj} and $d_{\tau j}$. The gravitational acceleration is denoted as g , and the unit constant vector is denoted as $e_3 = [0, 0, 1]^T$.

In the GNSS-denied environment without position information, the formation structure can be realized with leader-follower models based on visual feedback. To discuss the IBVS formation control problem, the leader-follower model based on visual feedback for quadrotors is depicted in Fig. 1. The follower quadrotor j is equipped with a forwarding-looking perspective camera, and uses the visual information to track the leader quadrotor i . To enhance the extraction of visual information from the leader quadrotor i , a sphere feature with radius r_i is mounted beneath it. The fixed relative position from the center of mass of the leader quadrotor i to the sphere feature is $\bar{h}_i = [0, 0, h_i]^T$. Let the camera frame ${}^c\mathcal{F}_j = \{{}^cX_j, {}^cY_j, {}^cZ_j\}$ overlap with the body frame ${}^b\mathcal{F}_j$ and introduce a virtual camera frame ${}^v\mathcal{F}_j = \{{}^vX_j, {}^vY_j, {}^vZ_j\}$ that inherits only the translational motion of the quadrotor, excluding rotational motion. Let $p_{ij} = p_i - p_j$ be the relative position between quadrotors i and j in ${}^c\mathcal{F}_j$, and the sphere feature on the leader quadrotor i in ${}^c\mathcal{F}_j$ can be denoted as

$${}^c p_{sij} = R_j^T (p_{ij} + \bar{h}_i). \quad (2)$$

The rotation matrix between ${}^c\mathcal{F}_j$ and ${}^v\mathcal{F}_j$ is R_j so that the coordinate of the leader quadrotor i in ${}^v\mathcal{F}_j$ can be given by

$${}^v p_{sij} = R_j {}^c p_{sij}$$

where ${}^v p_{sij}$ is the relative position between the sphere feature on the leader quadrotor i and the follower quadrotor j in ${}^v\mathcal{F}_j$. Therefore, the leader-follower dynamics can be derived by

$$\dot{{}^v p_{sij}} = v_i - v_j. \quad (3)$$

B. Image Dynamics

The sphere feature on the leader quadrotor i is denoted as ${}^v p_{sij} = [x_{ij}, y_{ij}, z_{ij} + h_i]^T$ in ${}^v\mathcal{F}_j$, and it can be captured by the virtual camera of the follower j . Based on the perspective

projection model, the image point on the virtual image plane can be given by

$$\begin{bmatrix} u_{ij} \\ n_{ij} \end{bmatrix} = \frac{\lambda_j}{x_{ij}} \begin{bmatrix} y_{ij} \\ z_{ij} + h_i \end{bmatrix}$$

where λ_j is the focal length of the perspective camera on the follower j . The dynamics of the image point are obtained by

$$\begin{bmatrix} \dot{u}_{ij} \\ \dot{n}_{ij} \end{bmatrix} = \begin{bmatrix} \frac{u_{ij}}{x_{ij}} & -\frac{\lambda_j}{x_{ij}} & 0 \\ \frac{n_{ij}}{x_{ij}} & 0 & -\frac{\lambda_j}{x_{ij}} \end{bmatrix} (v_j - v_i)$$

where x_{ij} is the depth information of the camera along the optical axis. Because the depth information can hardly be represented using the point feature of a perspective camera, a sphere-based image moment is introduced. Define the sphere-based image moment of order $p + q$ as

$$m_{ij}^{pq} = \iint u_{ij}^p n_{ij}^q du_{ij} dn_{ij}.$$

Then, the image feature $s_{ij} = [s_{xij}, s_{yij}, s_{zij}]^T \in \mathbb{R}^3$ of the leader quadrotor i and follower quadrotor j can be given by

$$s_{xij} = \sqrt{a_{ij}^* / a_{ij}}, \quad s_{yij} = s_{xij} u_{gij} / \lambda_j, \quad s_{zij} = s_{xij} n_{gij} / \lambda_j \quad (4)$$

where $u_{gij} = m_{ij}^{10} / a_{ij}$, $n_{gij} = m_{ij}^{01} / a_{ij}$, and $a_{ij} = m_{ij}^{00}$; (u_{gij}, n_{gij}) denotes the spherical center coordinate; a_{ij} is the area of the projected image; and a_{ij}^* is the corresponding desired value. By taking the derivative of (4), the image dynamics can be obtained by

$$\dot{s}_{ij} = v_{ij} / x_{ij}^* \quad (5)$$

where $v_{ij} = v_i - v_j$ is the relative linear velocity and x_{ij}^* is the desired displacement between quadrotors i and j . By extracting the spherical feature of the leader quadrotor and solving for the image moment features of all points on the sphere's surface, the visual features can denote the 3-D relative position.

C. Problem Formulation

To formulate the IBVS formation control system, it is necessary to combine the leader-follower dynamics that describe the relative motion between quadrotors, with the image dynamics derived from visual feedback. Let the image feature error be

$$e_{ij} = s_{ij} - s_{ij}^*$$

where s_{ij}^* is the desired image feature. The desired image feature is related to the desired relative position between the sphere feature on the leader quadrotor i and the follower quadrotor j and can be calculated from the desired formation structure. From (1) and (5), the leader-follower image error dynamics of quadrotors i and j can be given by

$$\begin{aligned} \dot{e}_{ij} &= v_{ij} / x_{ij}^* \\ \dot{v}_{ij} &= -ge_3 + R_j e_3 f_{Tj} / m_j - d_{fj} + \dot{v}_i. \end{aligned} \quad (6)$$

The leader-follower structure serves as a foundational framework that can be extended to various formation configurations in multiple quadrotor systems. In this framework, each follower is assigned to track a specific leader. A global leader is defined as the quadrotor that moves independently without following any other quadrotor. The remaining quadrotors

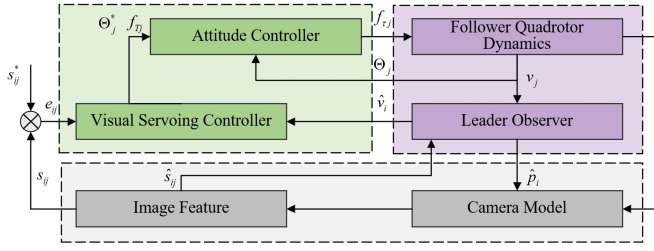


Fig. 2. Structure of IBVS formation control system.

maintain their desired relative positions with respect to their designated leaders, specifically by keeping the relative position between leader quadrotor i and follower quadrotor j at the desired value p_{ij}^* using visual feedback features. Because the follower quadrotor cannot obtain the state of the leader quadrotor, a leader observer is proposed in Section III. The control objective of this article is to derive the image feature error and relative linear velocity to zero (i.e., $e_{ij} \rightarrow 0$ and $v_{ij} \rightarrow 0$), such that the relative positions approach desired values and the follower velocity converges to that of the leader.

III. VISUAL SERVOING FORMATION PROTOCOL DESIGN

In this section, a leader observer is designed to estimate the leader state, while visual servoing and attitude control laws are formulated within the quadrotor dual-loop control framework. These components together constitute the closed-loop IBVS formation control system, as illustrated in Fig. 2.

A. Leader Observer Design

Define $x_{p0} = [p_0^T, v_0^T]^T \in \mathbb{R}^6$ as the global leader state for determining the formation trajectories and $x_{pi} = [p_i^T, v_i^T]^T$ as the state vector of the leader quadrotor i . Let p_{0i} denote the desired relative position between the leader quadrotor i and the global leader quadrotor. The leader quadrotor i can be denoted as $x_{pi} = x_{p0} + \tilde{p}_{0i}$, where $\tilde{p}_{0i} = [p_{0i}^T, 0_{1 \times 3}]^T$. It is assumed that $\|\dot{v}_i\| \leq \epsilon_i$ and $\|v_{ij}\| \leq \epsilon_{ij}$. Let $\hat{p}_i$ and $\hat{v}_i$ be the estimated values of the position and velocity of the leader quadrotor i , respectively, and $\hat{p}_{0i}$ be the estimated value of p_{0i} . The leader observer is designed as

$$\begin{aligned} \dot{\hat{p}}_i &= \hat{v}_i + \eta_1 (p_0 + \hat{p}_{0i} - \hat{p}_i) \\ \dot{\hat{v}}_i &= \eta_1 (v_0 - \hat{v}_i) + \eta_2 (v_j - \hat{v}_i) + \eta_3 (s_{ij} - \hat{s}_{ij}) \\ \dot{\hat{p}}_{0i} &= \eta_4 (p_{0i} - \hat{p}_{0i}) \end{aligned} \quad (7)$$

where η_1, η_2, η_3 , and η_4 are the observer gains. s_{ij} is the true image feature observed by the onboard camera of the follower quadrotor j , and $\hat{s}_{ij}$ is the estimated image feature calculated using the estimated leader position $\hat{p}_i$ via the camera model. Let the observer approximate errors be

$$\begin{aligned} \tilde{p}_i &= p_i - \hat{p}_i \\ \tilde{v}_i &= v_i - \hat{v}_i \\ \tilde{p}_{0i} &= p_{0i} - \hat{p}_{0i}. \end{aligned} \quad (8)$$

Theorem 1: The proposed leader observer (7) can guarantee that the approximate errors $\tilde{p}_i$, $\tilde{v}_i$, and $\tilde{p}_{0i}$ converge to zeros,

if the observer gains satisfies that $\eta_1 > 1$, $2\eta_4 > \eta_1$, and both η_2 and η_3 are sufficiently large.

Proof: By taking the derivative of (8), one has that

$$\begin{aligned} \dot{\tilde{p}}_i &= \tilde{v}_i - \eta_1 (\tilde{p}_i - \tilde{p}_{0i}) \\ \dot{\tilde{v}}_i &= \dot{v}_i - (\eta_1 + \eta_2) \tilde{v}_i - \eta_2 v_{ij} - \eta_3 (s_{ij} - \hat{s}_{ij}) \\ \dot{\tilde{p}}_{0i} &= -\eta_4 \tilde{p}_{0i}. \end{aligned} \quad (9)$$

The Lyapunov function is selected as follows:

$$\mathcal{L}_{pi} = 2\tilde{p}_i^T \tilde{p}_i + 2\tilde{v}_i^T \tilde{v}_i + 2\tilde{p}_{0i}^T \tilde{p}_{0i}. \quad (10)$$

The derivative of (10) can be given by

$$\dot{\mathcal{L}}_{pi} = \tilde{p}_i^T \dot{\tilde{p}}_i + \tilde{v}_i^T \dot{\tilde{v}}_i + \tilde{p}_{0i}^T \dot{\tilde{p}}_{0i}. \quad (11)$$

Substituting the error dynamics (9) into (11) and applying Young's inequality yields that

$$\begin{aligned} \dot{\mathcal{L}}_{pi} &= \tilde{p}_i^T \tilde{v}_i - \eta_1 \tilde{p}_i^T \tilde{p}_i + \eta_1 \tilde{p}_i^T \tilde{p}_{0i} - (\eta_1 + \eta_2) \tilde{v}_i^T \tilde{v}_i \\ &\quad + \tilde{v}_i^T \dot{v}_i - \eta_2 \tilde{v}_i^T v_{ij} - \eta_3 \tilde{v}_i^T \tilde{s}_{ij} - \eta_4 \tilde{p}_{0i}^T \tilde{p}_{0i} \\ &\leq (1/2)(1 - \eta_1) \|\tilde{p}_i\|^2 + (\epsilon_i - \eta_2 \epsilon_{ij} - \eta_3 \tilde{s}_{ij}) \|\tilde{v}_i\| \\ &\quad + (1/2 - \eta_1 - \eta_2) \|\tilde{v}_i\|^2 + ((1/2)\eta_1 - \eta_4) \|\tilde{p}_{0i}\|^2 \end{aligned} \quad (12)$$

where $\tilde{s}_{ij} = s_{ij} - \hat{s}_{ij}$ denotes the estimated image feature error introduced by the leader observer and $\tilde{s}_{ij}$ converges to zero as $\hat{p}_i \rightarrow p_i$. From (12), $\dot{\mathcal{L}}_{pi} < 0$ is satisfied if $\eta_1 > 1$, $2\eta_4 > \eta_1$, and both η_2 and η_3 are sufficiently large. Consequently, under these conditions, the observer errors asymptotically converge to zeros.

Algorithm 1 Off-Policy RL Algorithm for the Leader-Follower Visual Servoing System

- 1: **procedure**
- 2: Collect the state, control input, and disturbance input of the leader-follower visual servoing system for the quadrotor j at each time interval T , and record N_{sj} sets of system data with the fixed control input u_{sj} under the disturbance input d_{fj} .
- 3: Set the iteration index $k = 0$, and initialize admissible control and disturbance inputs u_{sj}^0, d_{fj}^0 .
- 4: **while** $\|u_{sj}^{k+1} - u_{sj}^k\| \geq \varrho_{su_j}$ or $\|d_{fj}^{k+1} - d_{fj}^k\| \geq \varrho_{sd_j}$, **do**
- 5: Solve the Bellman equation (20) for V_{sj}^k, u_{sj}^{k+1} , and d_{fj}^{k+1} , simultaneously, using the given control and disturbance inputs u_{sj}^k, d_{fj}^k .
- 6: $k \leftarrow k + 1$
- 7: **end while**
- 8: **end procedure**

B. Visual Servoing Control Law Design

Based on the approximate state of the leader quadrotor i obtained from the leader observer (7), the estimated relative velocity is denoted as $\hat{v}_{ij} = \hat{v}_i - v_j$. By substituting the dynamics of the estimated leader velocity into (6), one can have the leader-follower visual servoing dynamics as follows:

$$\begin{aligned} \dot{e}_{ij} &= \hat{v}_{ij} / x_{ij}^* \\ \dot{\hat{v}}_{ij} &= -ge_3 + R_j e_3 f_{Tj} / m_j - d_{fj} + o_{ij} \end{aligned} \quad (13)$$

where $o_{ij} = \eta_1(v_0 - \hat{v}_i) + \eta_2(v_j - \hat{v}_i) + \eta_3 \tilde{s}_{ij}$ is the related term introduced by the leader observer. The leader–follower visual servoing dynamics are used for the follower quadrotor j to design the visual servoing tracking control law, and o_{ij} is viewed as the time-varying term. Let $x_{sj} = [e_{ij}^T, \hat{v}_{ij}^T]^T \in \mathbb{R}^6$ be the state vector and $u_{sj} = R_j e_3 f_{Tj} \in \mathbb{R}^3$ be the virtual control input vector. The leader–follower visual servoing dynamics (13) can be rewritten as

$$\dot{x}_{sj} = F_{sj}x_{sj} + G_{sj}u_{sj} + C_{sj} + D_{sj}d_{fj} \quad (14)$$

where the system matrices are

$$F_{sj} = \begin{bmatrix} 0_{3 \times 3} & I_3/x_{ij}^* \\ 0_{3 \times 3} & 0_{3 \times 3} \end{bmatrix}, \quad G_{sj} = \begin{bmatrix} 0_{3 \times 3} \\ I_3/m_j \end{bmatrix}$$

$$C_{sj} = \begin{bmatrix} 0_{3 \times 3} \\ o_{ij} - g e_3 \end{bmatrix}, \quad D_{sj} = \begin{bmatrix} 0_{3 \times 3} \\ -I_3 \end{bmatrix}.$$

In order to derive the robust optimal control law for (14) based on the off-policy *RL*, the following disturbance attenuation criterion should be satisfied:

$$\int_t^\infty (x_{sj}^T \mathcal{Q}_{sj} x_{sj} + u_{sj}^T \mathcal{R}_{sj} u_{sj}) d\tau \leq \gamma_{sj}^2 \int_t^\infty \|d_{fj}\|^2 d\tau$$

where $\gamma_{sj} > 0$ is disturbance attenuation coefficient and $\mathcal{Q}_{sj}$ and $\mathcal{R}_{sj}$ are positive symmetric matrices. Then, the value function is defined as

$$V_{sj}(x_{sj}) = \int_t^\infty (x_{sj}^T \mathcal{Q}_{sj} x_{sj} + u_{sj}^T \mathcal{R}_{sj} u_{sj} - \gamma_{sj}^2 \|d_{fj}\|^2) d\tau.$$

By taking the derivative of the value function $V_{sj}(x_{sj})$ using Leibniz's formula, the Bellman equation is given by

$$0 = \nabla V_{sj}^T(x_{sj}) (F_{sj}x_{sj} + G_{sj}u_{sj} + C_{sj} + D_{sj}d_{fj})$$

$$+ x_{sj}^T \mathcal{Q}_{sj} x_{sj} + u_{sj}^T \mathcal{R}_{sj} u_{sj} - \gamma_{sj}^2 \|d_{fj}\|^2$$

$$\equiv H_{sj}(x_{sj}, \nabla V_{sj}, u_{sj}, d_{fj}). \quad (15)$$

By framing the robust optimal control problem as an H_∞ control problem, it can be solved using two-player zero-sum differential game theory, where the optimal value function $V_{sj}^*(x_{sj})$ satisfies

$$V_{sj}^*(x_{sj}) = \min_{u_{sj}} \max_{d_{fj}} \int_t^\infty (x_{sj}^T \mathcal{Q}_{sj} x_{sj} + u_{sj}^T \mathcal{R}_{sj} u_{sj}$$

$$- \gamma_{sj}^2 \|d_{fj}\|^2) d\tau.$$

The optimal control and disturbance inputs satisfy the stationarity conditions $\partial H_{sj}/\partial u_{sj} = 0$ and $\partial H_{sj}/\partial d_{fj} = 0$ at equilibrium. Therefore, one can obtain that

$$u_{sj}^*(x_{sj}) = -(1/2) \mathcal{R}_{sj}^{-1} G_{sj}^T \nabla V_{sj}^*(x_{sj})$$

$$d_{fj}^*(x_{sj}) = (1/2\gamma_{sj}^2) D_{sj}^T \nabla V_{sj}^*(x_{sj}). \quad (16)$$

The Hamilton–Jacobi–Isaacs (HJI) equation can be obtained as

$$0 = (\nabla V_{sj}^*)^T (F_{sj}x_{sj} + C_{sj}) + (1/4\gamma_{sj}^2) (\nabla V_{sj}^*)^T$$

$$\times D_{sj} D_{sj}^T \nabla V_{sj}^* + x_{sj}^T \mathcal{Q}_{sj} x_{sj}$$

$$- (1/4) (\nabla V_{sj}^*)^T G_{sj}^T \mathcal{R}_{sj}^{-1} G_{sj} \nabla V_{sj}^*. \quad (17)$$

It can be seen in (17) that in order to obtain the optimal control and disturbance inputs (16), one needs to solve the

HJI equation with completed system dynamics. Then, an off-policy *RL*-based control method is developed to approximate the solution of the HJI equation, enabling the derived robust model-free control law to overcome parameter uncertainties in the IBVS formation control system. The leader–follower visual servoing dynamics (14) can be rewritten as

$$\dot{x}_{sj} = F_{sj}x_{sj} + G_{sj}u_{sj}^k + C_{sj} + D_{sj}d_{fj}^k + G_{sj}(u_{sj} - u_{sj}^k)$$

$$+ D_{sj}(d_{fj} - d_{fj}^k) \quad (18)$$

where u_{sj}^k and d_{fj}^k are the evaluating policies to be improved in the k th iteration. Based on (15) and (16), by taking the derivative of the k th value function $V_{sj}^k(x_{sj})$ along with (18), one can have that

$$\dot{V}_{sj}^k(x_{sj}) = -x_{sj}^T \mathcal{Q}_{sj} x_{sj} - (u_{sj}^k)^T \mathcal{R}_{sj} u_{sj}^k + \gamma_{sj}^2 \|d_{fj}^k\|^2$$

$$- 2(u_{sj}^{k+1})^T \mathcal{R}_{sj} (u_{sj} - u_{sj}^k)$$

$$+ 2\gamma_{sj}^2 (d_{fj}^{k+1})^T (d_{fj} - d_{fj}^k). \quad (19)$$

Let T denote the integral time interval of the off-policy algorithm. The integral of (19) in $[t, t+T]$ can be given by

$$V_{sj}^k(x_{sj}(t+T)) - V_{sj}^k(x_{sj}(t))$$

$$= \int_t^{t+T} (-x_{sj}^T \mathcal{Q}_{sj} x_{sj} - (u_{sj}^k)^T \mathcal{R}_{sj} u_{sj}^k + \gamma_{sj}^2 \|d_{fj}^k\|^2) d\tau$$

$$+ \int_t^{t+T} (-2(u_{sj}^{k+1})^T \mathcal{R}_{sj} (u_{sj} - u_{sj}^k)) d\tau$$

$$+ \int_t^{t+T} (2\gamma_{sj}^2 (d_{fj}^{k+1})^T (d_{fj} - d_{fj}^k)) d\tau. \quad (20)$$

From (20), V_{sj}^k , u_{sj}^{k+1} , and d_{fj}^{k+1} can be solved simultaneously for a fixed control policy u_{sj} and a given disturbance d_{fj} . Algorithm 1 presents the off-policy *RL* framework for the leader–follower visual servoing system, enabling the derivation of the robust model-free visual servoing control law.

The implementation of Algorithm 1 requires neural networks to approximate the solution of (20). Specifically, three neural networks are employed to approximate the value function, the optimal control input, and the disturbance input as follows:

$$\hat{V}_{sj}^k(x_{sj}) = \hat{W}_{svj}^T \sigma_{svj}(x_{sj})$$

$$\hat{u}_{sj}^{k+1}(x_{sj}) = \hat{W}_{suj}^T \sigma_{suj}(x_{sj})$$

$$\hat{d}_{fj}^{k+1}(x_{sj}) = \hat{W}_{sdj}^T \sigma_{sdj}(x_{sj}) \quad (21)$$

where $\hat{V}_{sj}^k$, $\hat{u}_{sj}^{k+1}$, and $\hat{d}_{fj}^{k+1}$ are approximate values of V_{sj}^k , u_{sj}^{k+1} , and d_{fj}^{k+1} . $\hat{W}_{svj} \in \mathbb{R}^{l_{svj}}$, $\hat{W}_{suj} \in \mathbb{R}^{l_{suj} \times 3}$, and $\hat{W}_{sdj} \in \mathbb{R}^{l_{sdj} \times 3}$ are approximate weight vectors; $\sigma_{svj} \in \mathbb{R}^{l_{svj}}$, $\sigma_{suj} \in \mathbb{R}^{l_{suj}}$, and $\sigma_{sdj} \in \mathbb{R}^{l_{sdj}}$ are basis function vectors; and l_{svj} , l_{suj} , and l_{sdj} are numbers of neurons. Let $\hat{h}_{sj} = u_{sj} - u_{sj}^k$, $\hat{\lambda}_{sj} = d_{fj} - d_{fj}^k$, and $\mathcal{R}_{sj} = \text{diag}\{r_{sj,1}, r_{sj,2}, r_{sj,3}\}$. By substituting (21) into (20), the Bellman approximate error can be obtained as

$$\delta_{sj}(t) = \int_t^{t+T} (x_{sj}^T \mathcal{Q}_{sj} x_{sj} + (u_{sj}^k)^T \mathcal{R}_{sj} u_{sj}^k - \gamma_{sj}^2 \|d_{fj}^k\|^2) d\tau$$

$$+ 2 \sum_{\ell=1}^3 r_{sj,\ell} \int_t^{t+T} (\hat{W}_{suj,\ell}^T \sigma_{suj}(x_{sj}) \hat{h}_{sj,\ell}) d\tau$$

Algorithm 2 *RL-Based Robust Visual Servoing Control Algorithm for Quadrotors*

- 1: **Input:** Fixed control input u_{sj} , fixed disturbance input d_{fj} , integral time interval T , number of data samples N_{sj} , convergence threshold ϱ_{sj} , initial weights for the neural networks $\hat{W}_{svj}^0$, $\hat{W}_{su j}^0$, and $\hat{W}_{sdj}^0$.
- 2: **Output:** Approximate optimal control $\hat{u}_{sj}$ and optimal disturbance $\hat{d}_{fj}$ for the leader–follower visual servoing system
- 3: **Data Collection:** Collect the state, control input, and disturbance input of the leader–follower visual servoing system for quadrotor j at each time interval T , and record N_{sj} sets of system data with fixed u_{sj} under d_{fj} .
- 4: **Initialization:** Set the initial weights for the neural networks $\hat{W}_{svj}^k \leftarrow \hat{W}_{svj}^0$, $\hat{W}_{su j}^k \leftarrow \hat{W}_{su j}^0$, $\hat{W}_{sdj}^k \leftarrow \hat{W}_{sdj}^0$.
- 5: **while** $\|\hat{W}_{sj}^{k+1} - \hat{W}_{sj}^k\| \geq \varrho_{sj}$, $\forall x_{sj}$ **do**
- 6: Use collected data to compute Φ_{sj} and Y_{sj} .
- 7: Update approximate weights $\hat{W}_{sj}$ based on (24).
- 8: $k \leftarrow k + 1$
- 9: **end while**
- 10: Compute approximate optimal control and optimal disturbance for the leader–follower visual servoing system using the converged weights based on (21).
- 11: **Return:** $\hat{u}_{sj}$ and $\hat{d}_{fj}$

$$-2\gamma_{sj}^2 \sum_{\ell=1}^3 \int_t^{t+T} (\hat{W}_{sdj,\ell}^T \sigma_{sdj}(x_{sj}) \hat{\lambda}_{sj,\ell}) d\tau + \hat{W}_{svj}^T [\sigma_{svj}(x_{sj}(t+T)) - \sigma_{svj}(x_{sj}(t))] \quad (22)$$

where $\hat{W}_{su j,\ell}$ and $\hat{W}_{sdj,\ell}$ are the ℓ th column of $\hat{W}_{su j}$ and $\hat{W}_{sdj}$, respectively. The least-squares method is used to minimize the Bellman approximate error. By denoting $\varphi_{svj} = \sigma_{svj}(x_{sj}(t+T)) - \sigma_{svj}(x_{sj}(t))$, $\varphi_{su j,\ell} = 2r_{sj,\ell} \int_t^{t+T} (\sigma_{su j}(x_{sj}) \hat{h}_{sj,\ell}) d\tau$, and $\varphi_{sdj,\ell} = -2\gamma_{sj}^2 \int_t^{t+T} (\sigma_{sdj}(x_{sj}) \hat{\lambda}_{sj,\ell}) d\tau$ with $\ell = 1, 2, 3$, the Bellman approximate error (22) can be rewritten as

$$y_{sj}(t) + \delta_{sj}(t) = \hat{W}_{sj} \varphi_{sj}(t) \quad (23)$$

where $y_{sj}(t) = -\int_t^{t+T} (x_{sj}^T \mathcal{Q}_{sj} x_{sj} + (u_{sj}^k)^T \mathcal{R}_{sj} u_{sj}^k - \gamma_{sj}^2 \|d_{fj}^k\|^2) d\tau$, $\hat{W}_{sj} = [\hat{W}_{svj}^T, \hat{W}_{su j,1}^T, \hat{W}_{su j,2}^T, \hat{W}_{su j,3}^T, \hat{W}_{sdj,1}^T, \hat{W}_{sdj,2}^T, \hat{W}_{sdj,3}^T]^T$, and $\varphi_{sj}(t) = [\varphi_{svj}^T, \varphi_{su j,1}^T, \varphi_{su j,2}^T, \varphi_{su j,3}^T, \varphi_{sdj,1}^T, \varphi_{sdj,2}^T, \varphi_{sdj,3}^T]^T$. At the data collecting phase, by recording the system data at every time interval T from t_1 to $t_{N_{sj}}$, one has N_{sj} sets of data. Let $Y_{sj} = [y_{sj}(t_1), \dots, y_{sj}(t_{N_{sj}})]^T$ and $\Phi_{sj} = [\varphi_{sj}(t_1), \dots, \varphi_{sj}(t_{N_{sj}})]$. Assume that $N_{sj} \geq l_{svj} + 3l_{su j} + 3l_{sdj}$ is satisfied. The least-squares solution of (23) can be solved by

$$\hat{W}_{sj} = (\Phi_{sj} \Phi_{sj}^T)^{-1} \Phi_{sj} Y_{sj}. \quad (24)$$

Then, $\hat{V}_{sj}^k$, $\hat{u}_{sj}^{k+1}$, and $\hat{d}_{fj}^{k+1}$ can be solved by (24), and the detailed implementation procedure for solving the *RL*-based robust visual servoing control for the follower quadrotor j is displayed in Algorithm 2.

Once obtaining the outer loop visual servoing control $\hat{u}_{sj} = [u_{sj,x}, u_{sj,y}, u_{sj,z}]^T$, the thrust force and reference values for the inner loop attitude can be resolved. Denote the reference Euler

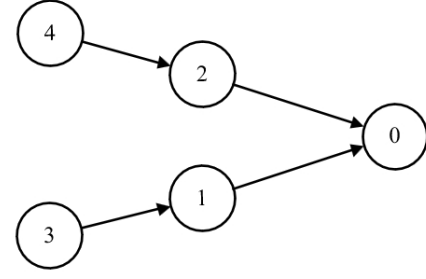


Fig. 3. Leader–follower formation topology in case studies.

angles as $\Theta_{rj} = [\phi_{rj}, \theta_{rj}, \psi_{rj}]^T$, and one can have that

$$\begin{aligned} f_{Tj} &= u_{sj,z} / (\cos \phi_{rj} \cos \theta_{rj}) \\ \phi_{rj} &= \arcsin(u_{sj,x} \sin \psi_{rj} - u_{sj,y} \cos \psi_{rj}) \\ \theta_{rj} &= \arctan((u_{sj,x} \cos \psi_{rj} + u_{sj,y} \sin \psi_{rj}) / f_{Tj}) \end{aligned}$$

where the reference yaw angle is set as a fixed constant for the quadrotors to synchronize.

Algorithm 3 *Off-Policy RL Algorithm for the Augmented Attitude System*

- 1: **procedure**
- 2: Collect the state, control input, and disturbance input of the augmented attitude system for the quadrotor j at each time interval T , and record $N_{\Theta j}$ sets of system data with the fixed control input $u_{\Theta j}$ under the disturbance input $d_{\tau j}$.
- 3: Set the iteration index $k = 0$, and initialize admissible control and disturbance inputs $u_{\Theta j}^0$, $d_{\tau j}^0$.
- 4: **while** $\|u_{\Theta j}^{k+1} - u_{\Theta j}^k\| \geq \varrho_{u\Theta j}$ or $\|d_{\tau j}^{k+1} - d_{\tau j}^k\| \geq \varrho_{d\tau j}$, $\forall x_{\Theta j}$ **do**
- 5: Solve the Bellman equation (29) for $V_{\Theta j}^k$, $u_{\Theta j}^{k+1}$, and $d_{\tau j}^{k+1}$, simultaneously, using the given control and disturbance inputs $u_{\Theta j}^k$, $d_{\tau j}^k$.
- 6: $k \leftarrow k + 1$
- 7: **end while**
- 8: **end procedure**

C. Attitude Control Law Design

For each quadrotor, the attitude control law is designed based on the off-policy *RL* algorithm in this section. Let $\bar{x}_{\Theta j} = [\Theta_j^T, \omega_j^T]^T$ be the state vector and $u_{\Theta j} = f_{\tau j}$ be the control input vector of the quadrotor j . From the dynamics of the quadrotor (1), the attitude dynamics of the quadrotor j can be written as

$$\dot{\bar{x}}_{\Theta j} = \bar{F}_{\Theta j}(\bar{x}_{\Theta j}) + \bar{G}_{\Theta j} u_{\Theta j} + \bar{D}_{\Theta j} d_{\tau j} \quad (25)$$

where the system matrices are

$$\begin{aligned} \bar{F}_{\Theta j} &= \begin{bmatrix} \omega_j I_3 \\ -J_j^{-1} (\omega_j \times J_j \omega_j) \end{bmatrix} \\ \bar{G}_{\Theta j} &= \begin{bmatrix} 0_{3 \times 3} \\ J_j^{-1} \end{bmatrix}, \quad \bar{D}_{\Theta j} = \begin{bmatrix} 0_{3 \times 3} \\ I_3 \end{bmatrix}. \end{aligned}$$

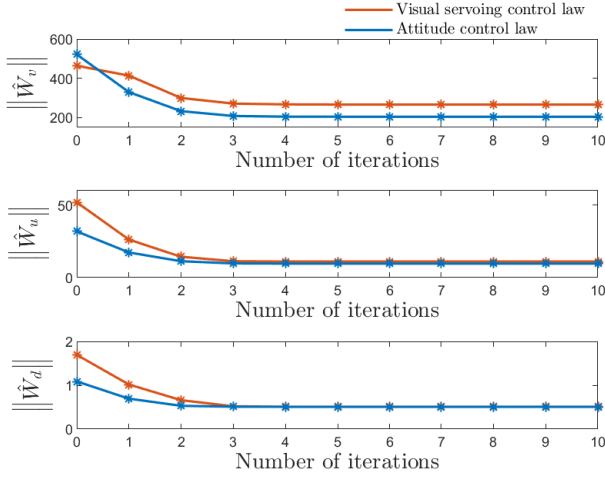


Fig. 4. Neural network weights of visual servoing and attitude control laws.

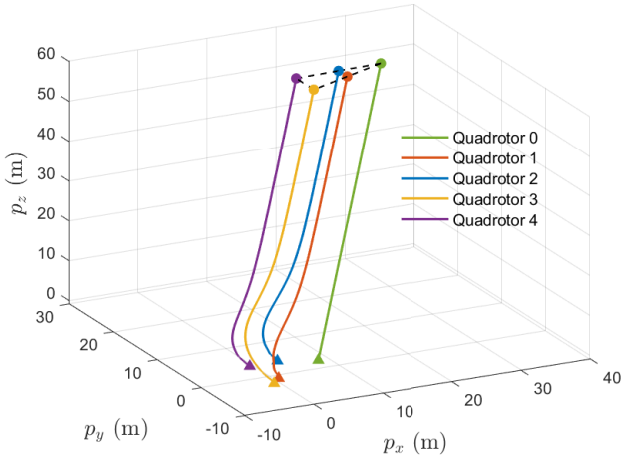
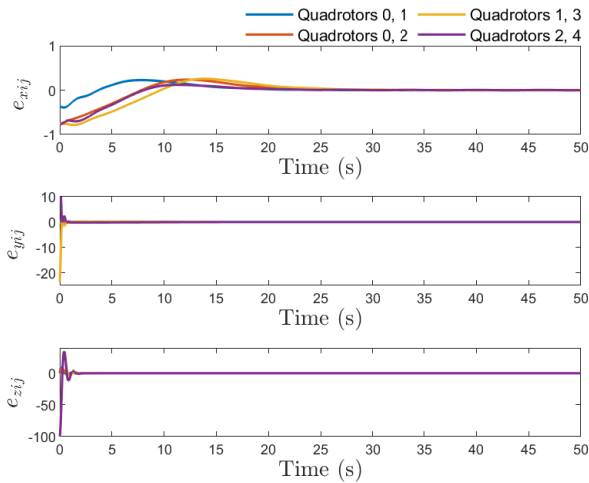
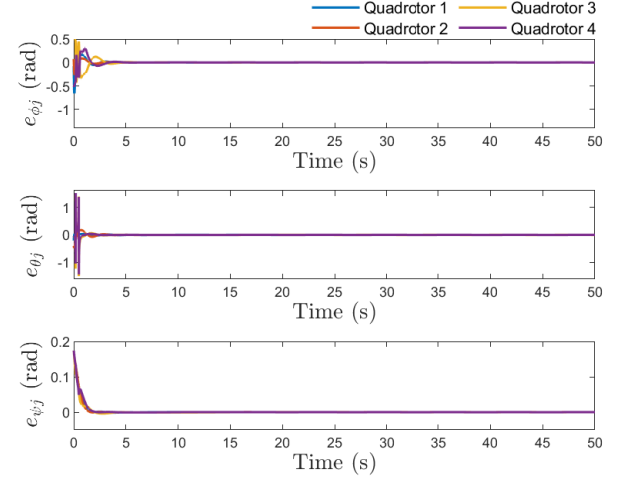
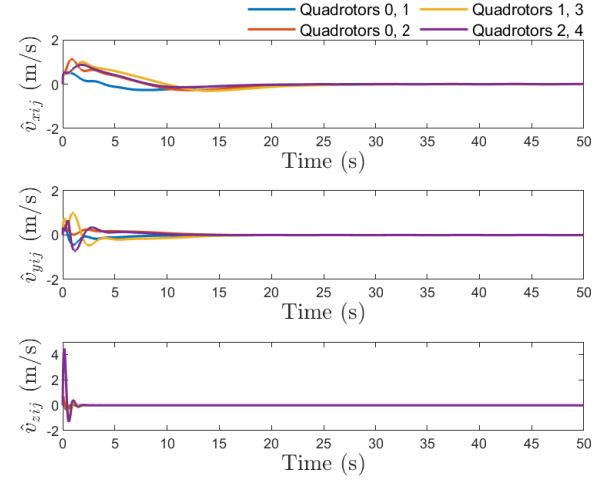
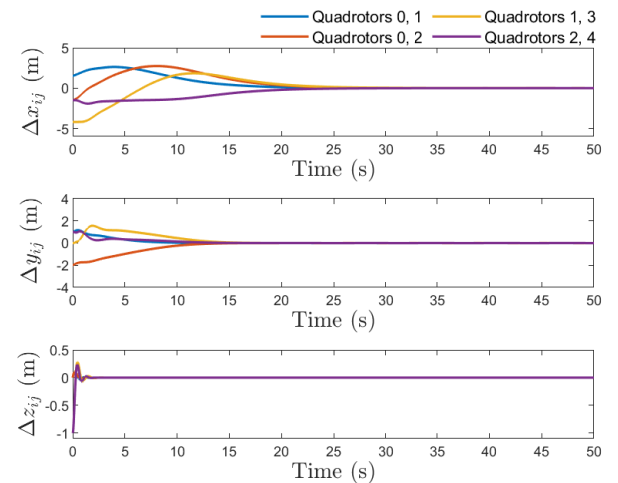


Fig. 5. Case 1: 3-D trajectories of IBVS formation system using the proposed method.

Fig. 6. Case 1: trajectories of the image feature errors e_{ij} between quadrotors i and j using the proposed method.

Let the reference attitude states of the quadrotor j be $\bar{x}_{\Theta rj} = [\Theta_{rj}^T, \omega_{rj}^T]^T$. The dynamics of the reference attitude are

Fig. 7. Case 1: trajectories of the attitude errors $e_{\theta j}$ for quadrotor j using the proposed method.Fig. 8. Case 1: trajectories of estimated relative linear velocities $\hat{v}_{ij}$ between quadrotors i and j using the proposed method.Fig. 9. Case 1: trajectories of relative position errors Δp_{ij} between quadrotors i and j using the proposed method.

described with an unknown smooth function $\bar{F}_{\Theta rj}(\bar{x}_{\Theta rj})$ as

$$\dot{\bar{x}}_{\Theta rj} = \bar{F}_{\Theta rj}(\bar{x}_{\Theta rj}). \quad (26)$$

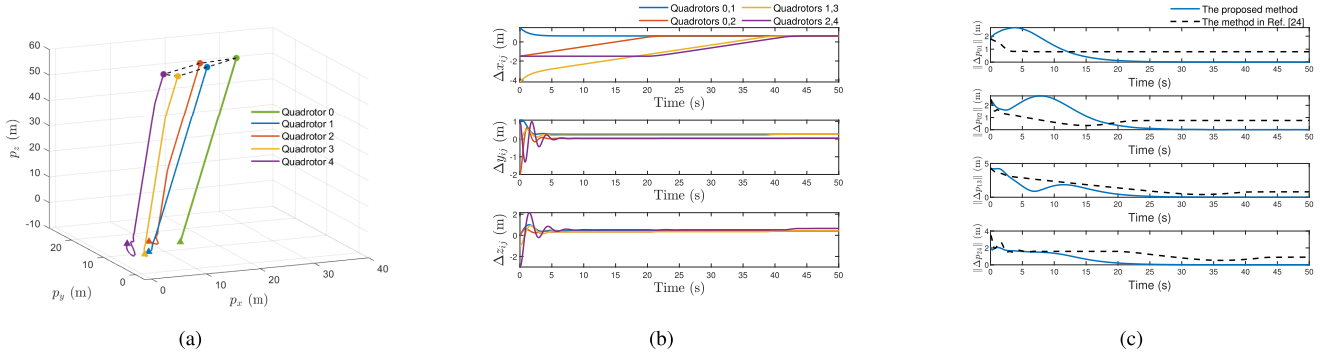


Fig. 10. Case 1: comparison results with the method in [24]. (a) 3-D trajectories using the method in [24]. (b) Relative position errors using the method in [24]. (c) Norm of the relative position errors: the proposed approach and method in [24].

Let the augmented attitude state be $x_{\Theta j} = [\bar{x}_{\Theta j}^T, \bar{x}_{\Theta r j}^T]^T \in \mathbb{R}^{12}$. From (25) and (26), the dynamics of the augmented attitude system can be given by

$$\dot{x}_{\Theta j} = F_{\Theta j}(x_{\Theta j}) + G_{\Theta j}u_{\Theta j} + D_{\Theta j}d_{\tau j} \quad (27)$$

where $F_{\Theta j}(x_{\Theta j}) = [\bar{F}_{\Theta j}^T(\bar{x}_{\Theta j}), \bar{F}_{\Theta r j}^T(\bar{x}_{\Theta r j})]^T$, $G_{\Theta j} = [\bar{G}_{\Theta j}^T, 0_{3 \times 6}]^T$, and $D_{\Theta j} = [\bar{D}_{\Theta j}^T, 0_{3 \times 6}]^T$. Let the attitude error be $e_{\Theta j} = \bar{x}_{\Theta j} - \bar{x}_{\Theta r j}$, and define the value function as

$$V_{\Theta j}(x_{\Theta j}) = \int_t^\infty (e_{\Theta j}^T Q_{\Theta j} e_{\Theta j} + u_{\Theta j}^T R_{\Theta j} u_{\Theta j} - \gamma_{\Theta j}^2 \|d_{\tau j}\|^2) d\tau$$

where $\gamma_{\Theta j} > 0$ is the disturbance attenuation coefficient and $Q_{\Theta j}$ and $R_{\Theta j}$ are positive symmetric matrices. The optimal control and disturbance inputs for the augmented attitude system can be given by

$$\begin{aligned} u_{\Theta j}^*(x_{\Theta j}) &= -(1/2) R_{\Theta j}^{-1} G_{\Theta j}^T \nabla V_{\Theta j}^*(x_{\Theta j}) \\ d_{\tau j}^*(x_{\Theta j}) &= (1/2\gamma_{\Theta j}^2) D_{\Theta j}^T \nabla V_{\Theta j}^*(x_{\Theta j}). \end{aligned}$$

The HJI equation of system (27) can be obtained as

$$\begin{aligned} 0 &= (\nabla V_{\Theta j}^*)^T F_{\Theta j}(x_{\Theta j}) - (1/4) (\nabla V_{\Theta j}^*)^T G_{\Theta j}^T R_{\Theta j}^{-1} G_{\Theta j} \nabla V_{\Theta j}^* \\ &\quad + e_{\Theta j}^T Q_{\Theta j} e_{\Theta j} + (1/4\gamma_{\Theta j}^2) (\nabla V_{\Theta j}^*)^T D_{\Theta j} D_{\Theta j}^T \nabla V_{\Theta j}^*. \end{aligned} \quad (28)$$

Similar to the control design for the leader–follower visual servoing system, the Bellman equation for the augmented attitude system (27) can be given by

$$\begin{aligned} &V_{\Theta j}^k(x_{\Theta j}(t+T)) - V_{\Theta j}^k(x_{\Theta j}(t)) \\ &= \int_t^{t+T} \left(-(e_{\Theta j}^k)^T Q_{\Theta j} e_{\Theta j}^k - (u_{\Theta j}^k)^T R_{\Theta j} u_{\Theta j}^k + \gamma_{\Theta j}^2 \|d_{\tau j}^k\|^2 \right) d\tau \\ &\quad + \int_t^{t+T} \left(-2(u_{\Theta j}^{k+1})^T R_{\Theta j} (u_{\Theta j}^k - u_{\Theta j}^{k+1}) \right) d\tau \\ &\quad + \int_t^{t+T} \left(2\gamma_{\Theta j}^2 (d_{\tau j}^{k+1})^T (d_{\tau j}^k - d_{\tau j}^{k+1}) \right) d\tau. \end{aligned} \quad (29)$$

In Algorithm 3, the off-policy RL framework for the augmented attitude system is developed, and the robust model-free attitude control law can be derived.

The implementation of Algorithm 3 follows a procedure similar to the leader–follower visual servoing system. Three neural networks are employed to approximate the value function, optimal control, and disturbance input. Using the least-squares method to solve the Bellman approximation error

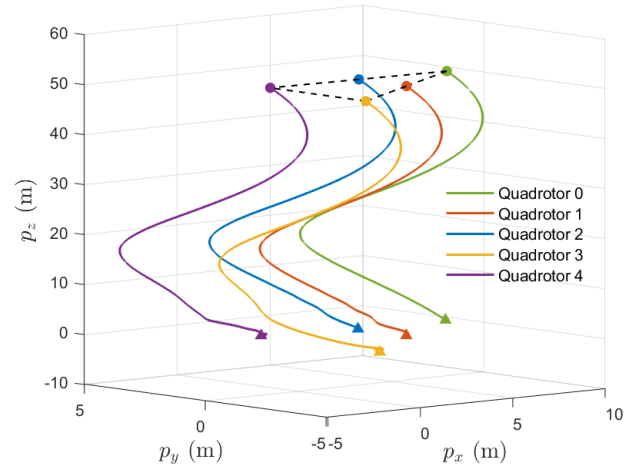


Fig. 11. Case 2: 3-D trajectories of IBVS formation system using the proposed method.

(27), the robust model-free attitude control law is ultimately obtained.

IV. CASE STUDY RESULTS

In this section, representative simulation studies are conducted to validate the effectiveness of the proposed leader observer, visual servoing control law, and attitude control law for the IBVS formation control system. Comparative studies with recent related methods are also provided to assess the performance of the proposed approach. The formation system involves five quadrotors in the case studies with the leader–follower interactive topology as shown in Fig. 3. Each quadrotor is modeled as in (1), equipped with a forward-looking perspective camera. The parameters of quadrotors and cameras are selected as: $m_1 = 1$ kg, $m_2 = 1.2$ kg, $m_3 = 1.6$ kg, $m_4 = 1.8$ kg, $J_j = \{0.0082, 0.0080, 0.0145\}$ kg · m²/rad², and $\lambda_j = 3.2$ mm with $j = 1, 2, 3, 4$. The uncertainties in the dynamics are simulated using $d_{fj} = 0.01[\sin t, \sin t, \sin t]^T$ and $d_{\tau j} = 0.05[\sin t, \sin t, \sin t]^T$. The sphere features are fixed below the leader quadrotors at a relative altitude $h_i = 1$ m and radius $r_i = 0.5$ m, where $i = 0, 1, 2$.

To evaluate the formation performance under different motion patterns, two cases are considered, including the IBVS

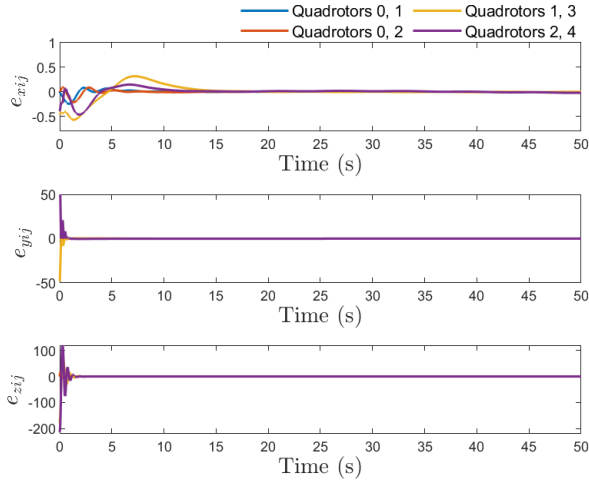


Fig. 12. Case 2: trajectories of the image feature errors e_{ij} between quadrotors i and j using the proposed method.

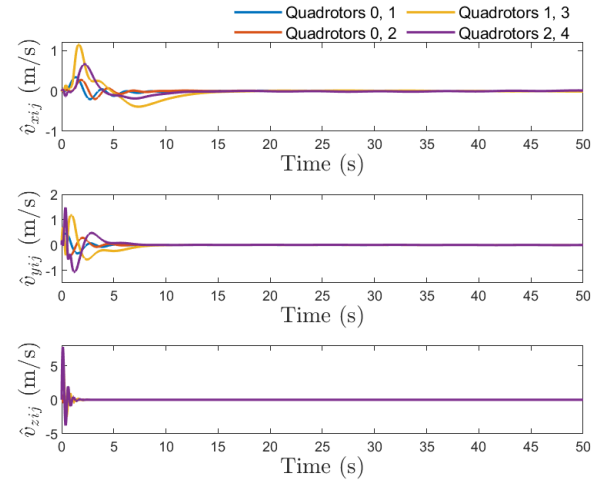


Fig. 14. Case 2: trajectories of estimated relative linear velocities $\hat{v}_{ij}$ between quadrotors i and j using the proposed method.

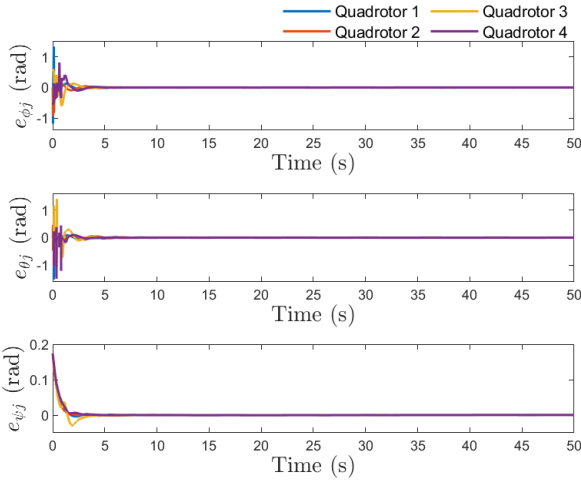


Fig. 13. Case 2: trajectories of the attitude errors $e_{\Theta j}$ for quadrotor j using the proposed method.

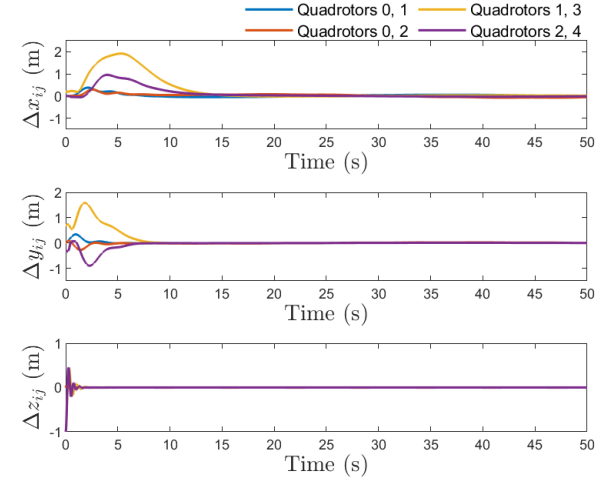


Fig. 15. Case 2: trajectories of relative position errors Δp_{ij} between quadrotors i and j using the proposed method.

formation tracking of a leader quadrotor following a line trajectory and a curved trajectory. For all leader–follower pairs, the observer gains are set as $\eta_1 = 50$, $\eta_2 = 5$, $\eta_3 = 5$, and $\eta_4 = 50$. The visual servoing and attitude control laws in Algorithms 1–3 are used to accomplish the formation mission, and the learning parameters are set as: $\mathcal{Q}_{sj} = 50I_6$, $\mathcal{R}_{sj} = I_3$, $\mathcal{Q}_{\Theta j} = 35I_6$, $\mathcal{R}_{\Theta j} = I_3$, $\gamma_{sj}^2 = 5$, and $\gamma_{\Theta j}^2 = 3$ with $j = 1, 2, 3, 4$. The integral time interval of the off-policy algorithm is set as $T = 0.05$ s, and the basis function vectors are chosen as polynomial functions of the system states.

Case 1: In this formation mission, the global leader moves along the trajectory $p_0 = [7 + 0.5t, 0.5t, 1 + t]^T$ m, and the follower quadrotors track the corresponding leader. Let the desired relative positions between the leaders and followers be $p_{01}^* = [5.5, 1, 1]^T$ m, $p_{02}^* = [5.5, -1, 1]^T$ m, $p_{13}^* = [5.5, 1, 1]^T$ m, and $p_{24}^* = [5.5, -1, 1]^T$ m. The initial positions of quadrotors are denoted as: $p_1 = [0, -2, 0]^T$ m, $p_2 = [3, 3, 0]^T$ m, $p_3 = [-1.3, -3, 0]^T$ m, and $p_4 = [-1, 5, 0]^T$ m. The initial velocities

and Euler angles of quadrotors are set as $[0, 0, 0]^T$ m/s and $[0, 0, 0.174]^T$ rad, respectively.

The results of Case 1 are displayed in Figs. 4–10. As shown in Fig. 4, the neural network weights of the visual servoing control law and attitude control law converge to the desired values by the fourth iteration, with a total training time of approximately 0.0266 s per episode in each training cycle.

Based on the proposed control laws, the trajectories of the quadrotors in the formation mission are depicted in Fig. 5. As shown in Figs. 6 and 7, the image feature errors and attitude errors converge to the desired values, validating the effectiveness of the proposed visual servoing and attitude control laws. Furthermore, the estimated relative linear velocities $\hat{v}_{ij}$ and relative position errors $\Delta p_{ij} = p_{ij} - p_{ij}^*$ of the quadrotors converge to ideal values, thus achieving the desired formation structure with visual feedback as shown in Figs. 8 and 9. To demonstrate the advantages of the proposed control method, the improved image error-based IBVS approach in [24] is applied to the same formation mission for comparison.

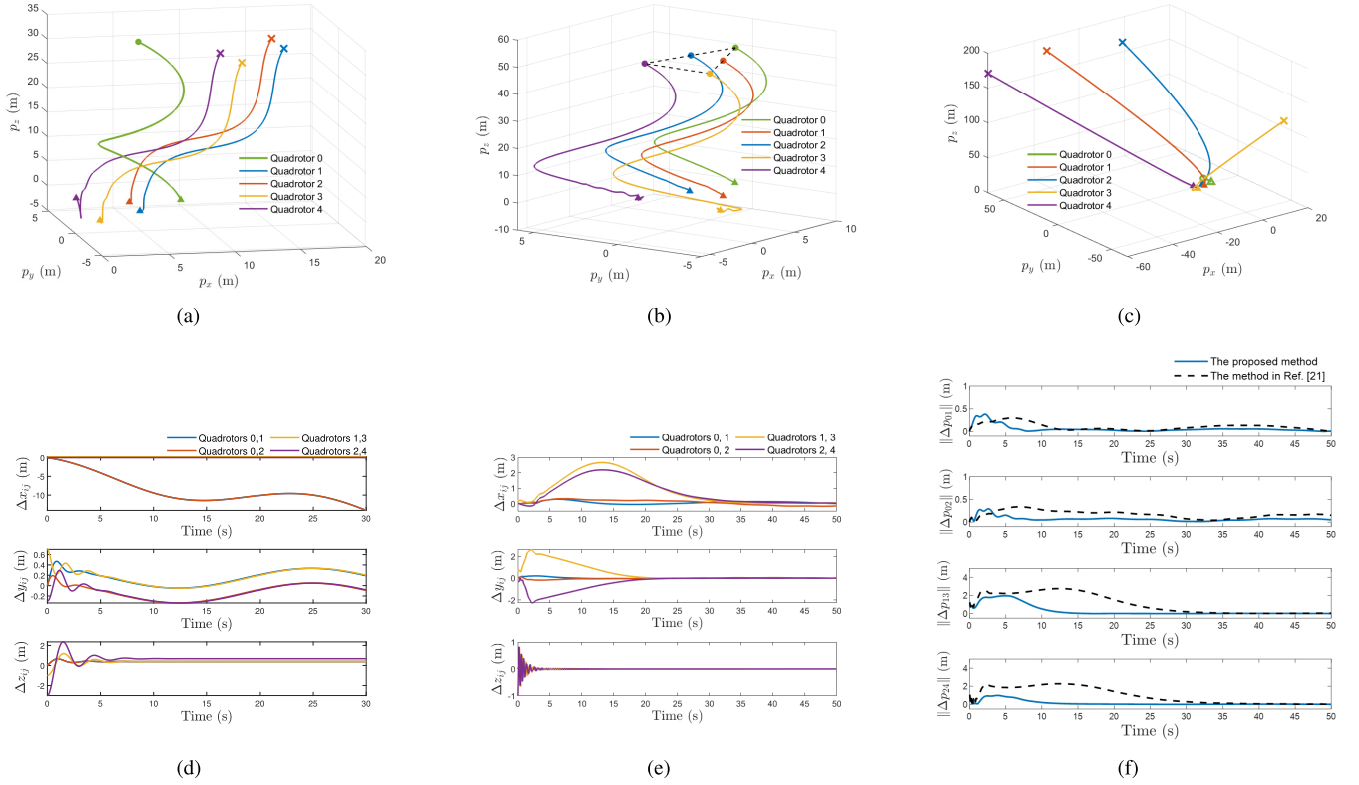


Fig. 16. Case 2: Comparison results with the methods in [21], [24], and [35]. (a) 3-D trajectories using the method in [24]. (b) 3-D trajectories using the method in [21]. (c) 3-D trajectories using the method in [35]. (d) Relative position errors using the method in [24]. (e) Relative position errors using the method in [21]. (f) Norm of the relative position errors: the proposed approach and method in [21].

As shown in Fig. 10, the method in [24] achieves basic formation visual servoing, but exhibits noticeable steady-state errors. The norm of the relative positions after convergence is 0.0030, 0.0039, 0.0008, and 0.0011 m with the proposed method, compared to 0.8032, 0.7576, 0.8031, and 0.9051 m using the method in [24] for the four follower quadrotors. These results indicate a significant improvement in steady-state formation accuracy achieved by the proposed approach.

Case 2: In this case, the global leader moves along the trajectory $p_0 = [5 + 3 \cos(2\pi t/25), 3 \sin(2\pi t/25), 1 + t]^T$ m. The desired relative positions between the leaders and followers are set as $p_{01}^* = [3.5, 1, 1]^T$ m, $p_{02}^* = [3.5, 1, 1]^T$ m, $p_{13}^* = [3.5, 1, 1]^T$ m, and $p_{24}^* = [3.5, -1, 1]^T$ m. The initial positions of quadrotors are denoted as: $p_1 = [4.5, -1, 0]^T$ m, $p_2 = [4.5, 1, 0]^T$ m, $p_3 = [0.8, -2.7, 0]^T$ m, and $p_4 = [1, 2.3, 0]^T$ m. The initial velocities and Euler angles of quadrotors are set as $[0, 0, 0]^T$ m/s and $[0, 0, 0.174]^T$ rad, respectively.

The results of Case 2 are illustrated in Figs. 11–16. The proposed control laws are still constructed with the converged neural network weights shown in Fig. 4. The trajectories of the quadrotors are depicted in Fig. 11. As shown in Figs. 12 and 13, the image feature errors and attitude errors converge to their desired values. Moreover, the estimated relative linear velocities and relative position errors converge as illustrated in Figs. 14 and 15, indicating that the desired formation structure is successfully achieved using visual feedback. In this case, three representative methods are selected for comparison, including the improved image error-based

IBVS control method in [24], the adaptive control method in [21], and the critic-actor-based RL control method in [35]. As shown in Fig. 16(a) and (d), the improved image error-based IBVS method in [24] fails to accomplish the formation mission, indicating its sensitivity to complex leader trajectories and formation constraints. In contrast, the proposed method exhibits enhanced robustness and reliability under challenging flight conditions. As shown in Fig. 16(b), the adaptive control method in [21] completes the formation task, but exhibits transient oscillations and steady-state errors, as illustrated in Fig. 16(e) and (f). After convergence, the norm of the relative position errors for the four quadrotors are 0.0089, 0.0507, 0.0121, and 0.0115 m under the proposed method, compared with 0.0020, 0.1498, 0.0559, and 0.0240 m using the method in [21], demonstrating the reduced overshoot and improved steady-state accuracy of the proposed approach. Furthermore, as shown in Fig. 16(c), servoing failure occurs when applying the method in [35] to this IBVS formation task. This failure is mainly attributed to the online neural network updates, limiting the real-time visual servoing performance. In contrast, the proposed off-policy RL-based control laws avoid this issue and are more suitable for real-time IBVS formation control.

V. CONCLUSION

In this article, the robust IBVS formation control problem is addressed for quadrotors in GNSS-denied and communication-degraded environments. The leader–follower image error dynamics are derived with unknown system uncertainties to

model the formation control problem. The leader observer is designed to estimate the unknown leader state, enabling formation control without intervehicle communication. The robust visual servoing and attitude control laws are derived using the RL algorithm to maintain formation and compensate for system uncertainties. The theoretical analysis and case study results are provided to verify the effectiveness of the proposed control framework. Although the current study demonstrates promising results in the case study, certain real-world factors, such as sensing noise and actuator limitations, remain to be explored. In the future, these limitations will be systematically considered, and the cooperative control problem without communication for the heterogeneous unmanned systems will be further investigated.

REFERENCES

- [1] Z. Wang, Y. Zou, Y. Liu, and Z. Meng, "Distributed control algorithm for leader-follower formation tracking of multiple quadrotors: Theory and experiment," *IEEE/ASME Trans. Mechatronics*, vol. 26, no. 2, pp. 1095–1105, Apr. 2021.
- [2] Y. Xiang, S. Li, R. Li, Z. Zhao, and H. Zhang, "Decentralized consensus inference-based hierarchical reinforcement learning for multiconstrained UAV pursuit-evasion game," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 36, no. 10, pp. 18229–18243, Oct. 2025.
- [3] O. Tutsoy, D. Asadi, K. Ahmadi, S. Y. Nabavi-Chashmi, and J. Iqbal, "Minimum distance and minimum time optimal path planning with bioinspired machine learning algorithms for faulty unmanned air vehicles," *IEEE Trans. Intell. Transp. Syst.*, vol. 25, no. 8, pp. 9069–9077, Aug. 2024.
- [4] Y. Wang, W. Liu, J. Liu, and C. Sun, "Cooperative USV-UAV marine search and rescue with visual navigation and reinforcement learning-based control," *ISA Trans.*, vol. 137, pp. 222–235, Jun. 2023.
- [5] K.-K. Oh, M.-C. Park, and H.-S. Ahn, "A survey of multi-agent formation control," *Automatica*, vol. 53, pp. 424–440, Mar. 2015.
- [6] Z. Yu, J. Li, Y. Xu, Y. Zhang, B. Jiang, and C.-Y. Su, "Reinforcement learning-based fractional-order adaptive fault-tolerant formation control of networked fixed-wing UAVs with prescribed performance," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 35, no. 3, pp. 3365–3379, Mar. 2024.
- [7] M. Cheng, H. Liu, G. Wen, J. Lü, and F. L. Lewis, "Data-driven time-varying formation-containment control for a heterogeneous air-ground vehicle team subject to active leaders and switching topologies," *Automatica*, vol. 153, Jul. 2023, Art. no. 111029.
- [8] X. Dong, Q. Wang, J. Yu, J. Lü, and Z. Ren, "Neuroadaptive output formation tracking for heterogeneous nonlinear multiagent systems with multiple nonidentical leaders," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 35, no. 3, pp. 3702–3712, Mar. 2024.
- [9] B. Deng, J. Nan, W. Cao, and W. Wang, "A survey on integration of network communication into vehicle real-time motion control," *IEEE Commun. Surveys Tuts.*, vol. 25, no. 4, pp. 2755–2790, 4th Quart., 2023.
- [10] H. Liu, J. Zhang, P. Zu, and M. Zhou, "Evolutionary algorithm-based attack strategy with swarm robots in denied environments," *IEEE Trans. Evol. Comput.*, vol. 27, no. 6, pp. 1562–1574, Dec. 2023.
- [11] T. Wang, Y. Zhao, J. Wang, A. K. Somani, and C. Sun, "Attention-based road registration for GPS-denied UAS navigation," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 32, no. 4, pp. 1788–1800, Apr. 2021.
- [12] J. Lin, Z. Miao, H. Zhong, W. Peng, Y. Wang, and R. Fierro, "Adaptive image-based leader-follower formation control of mobile robots with visibility constraints," *IEEE Trans. Ind. Electron.*, vol. 68, no. 7, pp. 6010–6019, Jul. 2021.
- [13] Z. Guo, J. Guo, J. Zhou, and J. Chang, "Robust tracking for hypersonic reentry vehicles via disturbance estimation-triggered control," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 56, no. 2, pp. 1279–1289, Apr. 2020.
- [14] R. Mebarki, V. Lippiello, and B. Siciliano, "Nonlinear visual control of unmanned aerial vehicles in GPS-denied environments," *IEEE Trans. Robot.*, vol. 31, no. 4, pp. 1004–1017, Aug. 2015.
- [15] D. Zheng, H. Wang, W. Chen, and Y. Wang, "Planning and tracking in image space for image-based visual servoing of a quadrotor," *IEEE Trans. Ind. Electron.*, vol. 65, no. 4, pp. 3376–3385, Apr. 2018.
- [16] S. Huang, H. Zeng, W. Lan, and X. Yu, "Leader-follower formation tracking control of mobile robots: A visual observer-based approach," *IEEE Trans. Control Syst. Technol.*, vol. 33, no. 5, pp. 1937–1945, Sep. 2025.
- [17] B. Luo, Y. Zhou, L. Liu, J. Xiao, B.-C. Wang, and C. Yang, "Reinforcement learning-based optimal formation control for multiple WMRs with visual servoing," *IEEE Trans. Autom. Sci. Eng.*, vol. 22, pp. 21284–21295, 2025.
- [18] N. Wang, H. He, Y. Hou, and B. Han, "Model-free visual servo swarming of manned-unmanned surface vehicles with visibility maintenance and collision avoidance," *IEEE Trans. Intell. Transp. Syst.*, vol. 25, no. 1, pp. 697–709, Jan. 2024.
- [19] Y. Chen et al., "Image-based visual servoing of unmanned aerial manipulators for tracking and grasping a moving target," *IEEE Trans. Ind. Informat.*, vol. 19, no. 8, pp. 8889–8899, Aug. 2023.
- [20] Y. Huang, M. Zhu, Z. Zheng, and K. H. Low, "Homography-based visual servoing for underactuated VTOL UAVs tracking a 6-DOF moving ship," *IEEE Trans. Veh. Technol.*, vol. 71, no. 3, pp. 2385–2398, Mar. 2022.
- [21] H. Liu, D. Liu, and Y. Lyu, "Completely distributed time-varying formation target tracking for quadrotor team via image-based visual servoing," *IEEE Trans. Veh. Technol.*, vol. 71, no. 1, pp. 21–32, Jan. 2022.
- [22] Y. Lyu, Q. Pan, C. Zhao, Y. Zhang, and J. Hu, "Vision-based UAV collision avoidance with 2D dynamic safety envelope," *IEEE Aerosp. Electron. Syst. Mag.*, vol. 31, no. 7, pp. 16–26, Jul. 2016.
- [23] P. Sun, B. Zhu, Z. Zuo, and M. V. Basin, "Vision-based finite-time uncooperative target tracking for UAV subject to actuator saturation," *Automatica*, vol. 130, Aug. 2021, Art. no. 109708.
- [24] G. Wang, J. Qin, Q. Liu, Q. Ma, and C. Zhang, "Image-based visual servoing of quadrotors to arbitrary flight targets," *IEEE Robot. Autom. Lett.*, vol. 8, no. 4, pp. 2022–2029, Apr. 2023.
- [25] M. Catellani and L. Sabattini, "Distributed control of a limited angular field-of-view multi-robot system in communication-denied scenarios: A probabilistic approach," *IEEE Robot. Autom. Lett.*, vol. 9, no. 1, pp. 739–746, Jan. 2024.
- [26] S. Zuo, Y. Song, F. L. Lewis, and A. Davoudi, "Adaptive output containment control of heterogeneous multi-agent systems with unknown leaders," *Automatica*, vol. 92, pp. 235–239, Jun. 2018.
- [27] Z. Ren, H. Liu, J. Xi, and Y. Zheng, "Data-driven self-triggered cooperative control in satellite formation flying under switching topologies subject to communication link faults," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 61, no. 4, pp. 9619–9629, Aug. 2025.
- [28] O. Tutsoy, D. Asadi, K. Ahmadi, and S.-Y. Nabavi-Chashmi, "Robust reduced order Thau observer with the adaptive fault estimator for the unmanned air vehicles," *IEEE Trans. Veh. Technol.*, vol. 72, no. 2, pp. 1601–1610, Feb. 2023.
- [29] C. Yan, L. Yan, Y. Lv, and Y. Xia, "Fully distributed event-triggered anti-windup consensus of heterogeneous systems with input saturation and an active leader," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 35, no. 9, pp. 12993–13004, Sep. 2024.
- [30] Z. Miao, H. Zhong, Y. Wang, H. Zhang, H. Tan, and R. Fierro, "Low-complexity leader-following formation control of mobile robots using only FOV-constrained visual feedback," *IEEE Trans. Ind. Informat.*, vol. 18, no. 7, pp. 4665–4673, Jul. 2022.
- [31] H. Wang, D. Guo, X. Liang, W. Chen, G. Hu, and K. K. Leang, "Adaptive vision-based leader-follower formation control of mobile robots," *IEEE Trans. Ind. Electron.*, vol. 64, no. 4, pp. 2893–2902, Apr. 2017.
- [32] L. Yang, Z. Liu, X. Zhang, X. Wang, and L. Shen, "Image-based distributed predictive visual servo control for cooperative tracking of multiple fixed-wing UAVs," *IEEE Robot. Autom. Lett.*, vol. 9, no. 9, pp. 7779–7786, Sep. 2024.
- [33] D. Wang, N. Gao, D. Liu, J. Li, and F. L. Lewis, "Recent progress in reinforcement learning and adaptive dynamic programming for advanced control applications," *IEEE/CAA J. Autom. Sinica*, vol. 11, no. 1, pp. 18–36, Jan. 2024.
- [34] X. Huo, H. R. Karimi, X. Zhao, B. Wang, and G. Zong, "Adaptive-critic design for decentralized event-triggered control of constrained nonlinear interconnected systems within an identifier-critic framework," *IEEE Trans. Cybern.*, vol. 52, no. 8, pp. 7478–7491, Aug. 2022.
- [35] J. Guan, Y.-X. Li, Z. Hou, and C. K. Ahn, "Reinforcement learning-based event-triggered adaptive fixed-time optimal formation control of multiple QAAVs," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 61, no. 5, pp. 11849–11864, Oct. 2025.

- [36] H. Modares, F. L. Lewis, and Z.-P. Jiang, " H_∞ tracking control of completely unknown continuous-time systems via off-policy reinforcement learning," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 26, no. 10, pp. 2550–2562, Oct. 2015.
- [37] B. Luo, H.-N. Wu, and T. Huang, "Off-policy reinforcement learning for H_∞ control design," *IEEE Trans. Cybern.*, vol. 45, no. 1, pp. 65–76, Jan. 2015.
- [38] S. Li, P. Durdevic, and Z. Yang, "Trajectory tracking of underactuated VTOL aerial vehicles with unknown system parameters via IRL," *IEEE Trans. Autom. Control*, vol. 67, no. 6, pp. 3043–3050, Jun. 2022.
- [39] M. Cheng, H. Liu, J. Xi, and Y. Zheng, "Data-driven robust optimal guidance with input saturation via differential graphical game strategy for cooperative aerial vehicles," *IEEE Trans. Intell. Transp. Syst.*, vol. 26, no. 6, pp. 8612–8621, Jun. 2025.
- [40] W. Zhao, H. Liu, and F. L. Lewis, "Robust formation control for cooperative underactuated quadrotors via reinforcement learning," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 32, no. 10, pp. 4577–4587, Oct. 2021.
- [41] H. Liu, Z. Ren, H. Duan, and M. V. Basin, "Reinforcement learning-based formation control for networked fixed-wing UAVs: Self-triggered observer–feedforward–feedback design and experiment," *IEEE Trans. Syst., Man, Cybern., Syst.*, vol. 56, no. 2, pp. 1226–1236, Feb. 2026, doi: [10.1109/TSMC.2025.3646484](https://doi.org/10.1109/TSMC.2025.3646484).
- [42] F. Morbidi, G. L. Mariottini, and D. Prattichizzo, "Observer design via immersion and invariance for vision-based leader–follower formation control," *Automatica*, vol. 46, no. 1, pp. 148–154, Jan. 2010.



Xinning Yi received the M.S. degree in control science and engineering from Central South University, Changsha, China, in 2023. She is currently pursuing the Ph.D. degree with the Institute of Artificial Intelligence, Beihang University, Beijing, China.

Her current research interests include visual servoing control, quadrotor flight control, and reinforcement learning.



Hao Liu (Senior Member, IEEE) received the B.E. degree in control science and engineering from the Northwestern Polytechnical University, Xi'an, China, in 2008, and the Ph.D. degree in automatic control from Tsinghua University, Beijing, China, in 2013.

In 2012, he was a Visiting Student with the Research School of Engineering, The Australian National University, Canberra, ACT, Australia. Since 2013, he has been with the School of Astronautics, Beihang University, Beijing, where he is currently a Professor. From 2017 to 2018, he was a Visiting Scholar with the University of Texas at Arlington Research Institute, Fort Worth, TX, USA. Since 2020, he has been with the Institute of Artificial Intelligence, Beihang University. Since 2022, he has also been with the Zhongguancun Laboratory, Beijing. His research interests include formation control, reinforcement learning, robust control, nonlinear control, unmanned aerial vehicles, unmanned underwater vehicles, and multiagent systems.

Dr. Liu received the Best Paper Award on IEEE ICCA 2018. He serves as an Associate Editor for *International Journal of Robust and Nonlinear Control* and *Journal of Intelligent and Robotic Systems*.



Haibin Duan (Senior Member, IEEE) received the Ph.D. degree in control theory and engineering from Nanjing University of Aeronautics and Astronautics (NUAA), Nanjing, China, in 2005.

He is currently a Full Professor with the School of Automation Science and Electrical Engineering, Beihang University, Beijing, China, where he is the Head of the Bio-Inspired Autonomous Flight Systems (BAFS) Research Group. He has authored or co-authored more than 90 publications. His current research interests include multi-UAV swarm

autonomous control, bioinspired intelligence, and biological computer vision.

Dr. Duan received the National Science Fund for Distinguished Young Scholars of China in 2014. He is the Editor-in-Chief of *Guidance, Navigation and Control*; the Deputy Editor-in-Chief of *Acta Automatica Sinica*; and an Associate Editor of IEEE TRANSACTIONS ON CYBERNETICS, IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS: REGULAR PAPERS, and IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS—I: EXPRESS BRIEFS.



Biao Luo (Senior Member, IEEE) received the Ph.D. degree from Beihang University, Beijing, China, in 2014.

He was an Associate Professor and an Assistant Professor with the Institute of Automation, Chinese Academy of Sciences, Beijing, from 2014 to 2018. He is currently a Professor with the School of Automation, Central South University (CSU), Changsha, China. His current research interests include intelligent control, reinforcement learning, deep learning, and decision-making.

Dr. Luo is an Associate Editor of IEEE TRANSACTIONS ON NEURAL NETWORKS AND LEARNING SYSTEMS, *Artificial Intelligence Review*, and the *Neurocomputing*. He is the Vice Chair of the Adaptive Dynamic Programming and Reinforcement Learning Technical Committee, Chinese Association of Automation.



Shan Zuo (Member, IEEE) received the B.S. degree in electronic information science and control science and engineering and the Ph.D. degree in electrical engineering from the University of Electronic Science and Technology of China, Chengdu, China, in 2012 and 2018, respectively, and the second Ph.D. degree in electrical engineering from the University of Texas at Arlington, Arlington, TX, USA, in 2020.

She was a Full Professor with the College of Automation and the College of Artificial Intelligence, Nanjing University of Posts and Telecommunications, Nanjing, China. She is currently an Assistant Professor with the Department of Electrical and Computer Engineering, University of Connecticut (UConn), Storrs, CT, USA. Her research interests include distributed control, reinforcement learning, microgrids, autonomous systems, and cyber-physical systems.